A planar triangulation with no partition into two induced pseudoforests

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Abstract

We answer a question of L. Ploscaru (20th Emléktábla Workshop, 2026) in the negative: there is a planar triangulation on 58 vertices that admits no orientation with $\min(d^+(v), d^-(v)) \leq 1$ at every vertex v — equivalently, whose vertex set cannot be partitioned into two induced pseudoforests. The construction substitutes three copies of a non-Hamiltonian cubic fragment into K_4; the fragment is obtained by deleting a vertex from a smallest counterexample to Tait's conjecture, the 38-vertex Barnette–Bosák–Lederberg graph. Infeasibility follows from an Euler-characteristic identity for the system of dual curves induced by any candidate partition, together with a pigeonhole argument in the spirit of Tutte's refutation of Tait's conjecture. Exhaustive SAT verification shows that every planar triangulation on $n \leq 17$ vertices does admit such a partition, so a minimum counterexample has between 18 and 58 vertices.

1 Introduction

A *pseudoforest* is a graph in which every connected component contains at most one cycle. At the 20th Emléktábla Workshop (2026), L. Ploscaru posed the following problem [1, p. 17].

> **Problem 1.** Does every planar triangulation admit an orientation such that $\min\bigl(d^+(v), d^-(v)\bigr) \leq 1$ for every vertex v?

As noted there, requiring $\min = 0$ is impossible (it forces bipartiteness), while $\min \leq 2$ follows by an easy induction on a vertex of degree at most 5; and the problem is equivalent to asking whether $V(T)$ can be partitioned into two sets each inducing a pseudoforest. The two-*forest* analogue fails: planar graphs have vertex arboricity at most 3 [2], and the bound is sharp [3]; indeed a triangulation has vertex arboricity at most 2 if and only if its dual is Hamiltonian [5, 6], so the dual of any non-Hamiltonian 3-connected cubic planar graph — e.g. the 21-vertex duals of the 38-vertex graphs of [7] — is a triangulation needing three forests. Pseudoforests sit strictly between: a triangulation has $3n-6$ edges, and two induced pseudoforests can carry at most n of them, so the question is whether the remaining $\geq 2n-6$ edges can always be pushed into the bipartite cut — tantalizingly close to the $2n-4$ maximum for planar bipartite graphs.

We show that the answer is no.

Theorem A. There is a planar triangulation T^* on 58 vertices whose vertex set cannot be partitioned into two induced pseudoforests; equivalently, T^* has no orientation with $\min(d^+(v), d^-(v)) \leq 1$ everywhere.

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Theorem B. Every planar triangulation on $n \le 17$ vertices admits such a partition. Hence a minimum counterexample has $18 \le n \le 58$ vertices.

Here and throughout, a *planar triangulation* is a simple maximal planar graph on $n \ge 4$ vertices (automatically 3-connected); this is the class generated by `plantri` [12] and enumerated by OEIS A000109.

The equivalence between the orientation and partition formulations is recalled in Proposition 2.1. Our main tool (Lemma 2.6) converts a candidate partition into a system of disjoint closed curves in the dual cubic graph and computes the cyclomatic number of each monochromatic piece from the curve topology. The counterexample then emerges from a Hamiltonicity obstruction: the curve system would have to thread a "through-Hamiltonian path" across non-Hamiltonian fragments, or pay for its failure in a currency (regions of excess rank) that the sphere cannot supply.

2 Partitions as curve systems

Throughout, T is a planar triangulation embedded in the sphere S^2, and $G = D(T)$ its dual: a 3-connected simple cubic planar graph, embedded so that each dual vertex lies in the interior of its triangle and each dual edge crosses its primal edge exactly once and meets no other primal edge. Whitney's theorem makes these embeddings unique up to reflection, so the objects below are well defined.

Proposition 2.1 (equivalence; stated in [1])**.** *A graph G_0 has an orientation with* $\min(d^+(v), d^-(v)) \le 1$ *for all v if and only if $V(G_0)$ can be partitioned into two sets each inducing a pseudoforest.*

Proof. ($\Leftarrow$) Inside A, orient each pseudoforest component so that every vertex has out-degree at most 1: in a tree component, orient all edges toward a root; in a unicyclic component, orient the cycle cyclically and the remaining edges toward the cycle. Inside B, do the mirror image, so that every vertex has in-degree at most 1. Orient every cut edge from its B-endpoint to its A-endpoint: this adds only in-degree at A-vertices (which watch their out-degree) and only out-degree at B-vertices (which watch their in-degree). Then $\min(d^+, d^-) \le 1$ holds at every vertex.

($\Rightarrow$) Given the orientation, put $A = \{v : d^+(v) \le 1\}$ and $B = V \setminus A$; every $v \in B$ has $d^-(v) \le 1$. Inside A, every vertex has out-degree at most 1, so each component K of $G_0[A]$ has $e(K) \le |K|$: a pseudoforest. Symmetrically for B. $\square$

Fix now a partition $V(T) = A \cup B$. Let X be the set of bichromatic ("cut") edges and H the set of their dual edges in G.

Lemma 2.2 (curves)**.** *Every vertex of G has H-degree 0 or 2. Hence H is a disjoint union of simple cycles $C_1, \dots, C_k$ of G, drawn in S^2 as k pairwise disjoint simple closed curves. The union of the curves meets the primal graph exactly in one interior point of each cut edge.*

Proof. A triangle with vertex colours from $\{A, B\}$ has exactly 0 or 2 bichromatic edges. A 2-regular graph is a disjoint union of simple cycles. The drawing statement is the choice of dual embedding. $\square$

An H-cycle C crosses the cut edges dual to its edges; orient C arbitrarily and speak of the *left* and *right* endpoint of each crossed edge.

Lemma 2.3 (sides)**.** *Let C be one of the curves, with crossed edges $e_1, \dots, e_L$ in cyclic order. All left endpoints of the e_j lie in one monochromatic component $K_{\mathrm{left}}(C)$, all right endpoints in one monochromatic component $K_{\mathrm{right}}(C)$, and these two components have different colours (in particular $K_{\mathrm{left}}(C) \ne K_{\mathrm{right}}(C)$).*

Proof. Consecutive crossed edges e_j, e_{j+1} are the two cut edges of a common triangle $f = xyz$, say $e_j = xy$ and $e_{j+1} = xz$, where x is the minority-colour vertex of f. As C passes through f it keeps x on one fixed side and $\{y, z\}$ on the other, and yz is an uncut edge. So consecutive left endpoints are either the same vertex or joined by an uncut edge; hence, by induction around C, all left endpoints lie in one monochromatic component, and likewise on the right. A crossed edge has one endpoint of each colour, so the two components carry different colours. □

Lemma 2.4 (spheres and trees; folklore, cf. [11])**.** *k pairwise disjoint simple closed curves cut S^2 into exactly $k+1$ open connected regions; each curve lies on the boundary of exactly 2 regions; and the bipartite incidence graph (regions vs. curves) is a tree.*

Proof. Induction on k. For $k = 0$ there is one region. A new curve γ is connected and disjoint from the old curves, so it lies inside one existing region R. All curves here are polygonal (drawings of cycles of a plane graph), so γ has an annular collar inside R meeting no curve; by the Jordan curve theorem γ splits S^2 into two sides, and via the collar, $R \setminus \gamma$ has exactly two components, one on each side, each with γ on its boundary; no other region is affected. Thus regions increase by one, and the incidence graph is the old one with the node R split in two and joined through the new edge γ (old subtrees attach to the side of γ they lie in): it stays a tree. □

We call the tree of Lemma 2.4 the *nesting tree*. For a region R let b_R denote its number of boundary curves (its degree in the nesting tree).

Lemma 2.5 (regions are monochromatic components)**.** *Every monochromatic component lies in exactly one region; this map is a bijection from monochromatic components to regions; and under it the two regions flanking a curve C contain precisely $K_{\mathrm{left}}(C)$ and $K_{\mathrm{right}}(C)$. Consequently the combinatorial incidence (components vs. curves given by Lemma 2.3) coincides with the topological nesting tree.*

Proof. Primal vertices and uncut edges avoid the curves (Lemma 2.2), so a monochromatic component, being connected, lies in one region. Every region contains a primal vertex: any point p of R lies in a triangle f; if no curve passes through f, all of closed f (hence a vertex) is in R; if the single arc of a curve through f separates it, each side contains at least one corner of f. In particular the map from components to regions is surjective. Next, consider the auxiliary multigraph I with nodes the monochromatic components and one edge per curve C joining $K_{\mathrm{left}}(C)$ to $K_{\mathrm{right}}(C)$. I is connected: T is connected, and traversing a cut edge uv moves between exactly the two components flanking its curve (u is a left or right endpoint of a crossed edge of that curve, so its component is K_{left} or K_{right}; same for v). A connected graph with k edges has at most $k+1$ nodes, so #components $\leq k+1 =$ #regions; with surjectivity this forces #components $= k+1$ and the map is a bijection. Then I is connected with $k+1$ nodes and k edges: a tree. Finally the left endpoints of C lie locally on the left of C, i.e. in the region flanking C on that side, so I maps onto the topological incidence tree node-by-node and edge-by-edge. □

A dual vertex not lying on any curve (equivalently, a monochromatic triangle of T) is called a *stray*. For a region R, let w_R be the number of strays in R.

Lemma 2.6 (region rank)**.** *Let K be a monochromatic component and R its region. Then the cyclomatic number of $T[K]$ satisfies*

$$\mu(K) \ := \ e(K) - |K| + 1 \ = \ b_R - 1 + w_R.$$

Proof. The closed region $\bar{R}$ is the sphere minus b_R open disks, so $\chi(\bar{R}) = 2 - b_R$. We exhibit a polygonal cell decomposition of $\bar{R}$ and count. Write L_i for the number of crossings of the i-th boundary curve of R ($i = 1, \dots, b_R$) and $L = \sum_i L_i$.

0-*cells*: the $|K|$ primal vertices in R, plus the L crossing points (one per crossed edge of a boundary curve).

1-*cells*: the $e(K)$ uncut edges inside R; the L half-edges (for each crossing point, the half of its cut edge from the crossing point to its endpoint in K — exactly one endpoint of each crossed edge lies on R's side); and the L curve arcs between consecutive crossing points (each such arc lies inside a single triangle).

2-*cells*: the w_R monochromatic triangles of R, plus, for each of the L boundary arcs, the piece of its (bichromatic) triangle on R's side; each piece is a disk (one is bounded by the arc and two half-edges, the other by the arc, two half-edges and one uncut edge), and the two pieces of one triangle lie in different regions because the two flanks of a curve are different regions (Lemma 2.5).

These cells exactly cover $\bar{R}$: any triangle meeting R either lies in R (monochromatic, all uncut, hence entirely in one region) or is crossed by a boundary curve of R. Therefore

$$\chi(\bar{R}) = (|K| + L) - (e(K) + L + L) + (w_R + L) = |K| - e(K) + w_R,$$

and equating with $2 - b_R$ gives $e(K) - |K| + 1 = b_R - 1 + w_R$. $\square$

As a sanity check: with $k = 0$ there is one region, $b = 0$, and $w = 2n - 4$ faces, so the formula gives $\mu = 2n-5 = (3n-6)-n+1$, the cyclomatic number of T itself. The identity has also been machine-checked on every 2-colouring of every triangulation with $n \leq 10$ (134,296 instances) and on more than 10^6 larger instances; see Section 5.

Corollary 2.7 (validity criterion)**.** *The partition (A, B) is a pseudoforest 2-partition if and only if every region R satisfies*

$$b_R - 1 + w_R \;\leq\; 1. \tag{$\star$}$$

If ($\star$) holds then: every region has $b_R \leq 2$, i.e. the nesting tree is a path; there are at most two strays; strays lie only in regions with $b_R = 1$, i.e. in the two end regions of the path, at most one stray per end region; and a region containing a stray borders exactly one curve and contains no second stray.

Proof. The criterion is Lemma 2.6 applied to every monochromatic component ("every component has $\mu \leq 1$"). Note first that $k \geq 1$ in any valid partition: $k = 0$ means no cut edges, so one colour class is empty and T itself would have to be a pseudoforest, but $\mu(T) = 2n-5 \geq 3$ for $n \geq 4$. Now if ($\star$) holds, $w \geq 0$ forces $b \leq 2$ everywhere (a path), each region has $b \geq 1$, and $w_R \leq 2 - b_R \leq 1$ with $w_R = 1$ only if $b_R = 1$; a path has exactly two ends. $\square$

3 The construction

Let Γ be a Barnette–Bosák–Lederberg graph: a non-Hamiltonian 3-connected cubic planar graph on 38 vertices. The minimum order of such graphs is 38 and is attained by exactly six graphs [7]; they arise from the work of Barnette, of Bosák [8], and of Lederberg in the mid-1960s, so 38 is the order of the smallest counterexamples to Tait's conjecture [10, 9, 7]. We use the copy archived as House of Graphs #954 [13]; its non-Hamiltonicity is also re-verified mechanically here (Section 5). Fix any vertex v_0 of Γ and let $F = \Gamma - v_0$, a fragment with 34 cubic vertices and 3 vertices of degree 2 (the former neighbours of v_0), which we call *ports*.

Definition (the graph G^*). Take K_4 with vertices h, b_1, b_2, b_3. Substitute for each b_i a copy F_i of F, attaching its three ports to the three K_4-edges at b_i: one port to the edge

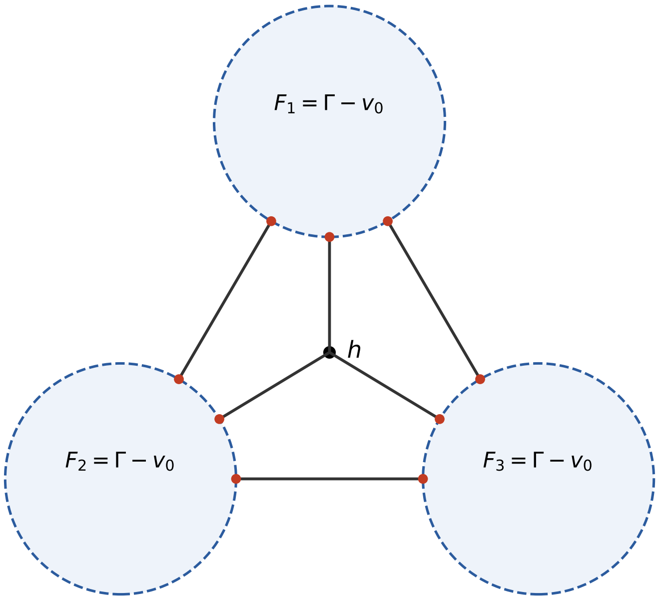

Figure 1: The cubic graph G^* on 112 vertices: a hub h and three copies F_i of $F = \Gamma - v_0$, wired as K_4. Ports (degree-2 vertices of the fragments) are shown as small dots; the six connector edges realize the K_4 pattern. The counterexample is $T^* = D(G^*)$.

toward h (the h-*port*) and one port each to the edges toward the other two fragments. Keep h as a single vertex. The resulting graph G^* is cubic on $3 \cdot 37 + 1 = 112$ vertices and planar (embed each F_i in a disk with its three ports on the boundary in the rotation inherited from a planar embedding of Γ around v_0). See Figure 1.

Lemma 3.1 (substitution preserves 3-connectivity)**.** *Let G_1, G_2 be 3-connected cubic graphs, $u \in V(G_1)$, $v \in V(G_2)$. Delete u and v and join the three former neighbours of u to the three former neighbours of v by a perfect matching. The resulting cubic graph G is 3-connected.*

Proof. Write $A_1 = V(G_1) - u$, $A_2 = V(G_2) - v$. Let $|S| \leq 2$ and suppose $G - S$ is disconnected. Note G_i minus one vertex is 2-connected (G_i is 3-connected), so if S lies entirely in A_1, then A_2 induces a connected subgraph of $G - S$. Any component C of $(G_1 - u) - S$ satisfies $N_{G_1}(C) \subseteq S \cup \{u\}$; if u is not adjacent to C then $N_{G_1}(C) \subseteq S$ is a cut of size at most 2 in G_1 (u lies outside C, so the complement of $C \cup S$ is nonempty), impossible; hence some vertex of C was a neighbour of u and retains its matching edge into A_2 in $G - S$. So every component of the A_1-side attaches to the connected A_2-side: $G - S$ is connected, a contradiction. If S has one vertex on each side, then each of $(G_1 - u) - s_1$ and $(G_2 - v) - s_2$ is connected (2-connected minus one vertex), and each deleted vertex destroys at most one of the three matching edges, so a matching edge survives and joins the two sides. The cases $S \subseteq A_2$ and $|S| \leq 1$ are symmetric or easier. □

Applying Lemma 3.1 three times (substituting Γ for each of the three chosen vertices of K_4; both K_4 and Γ are 3-connected cubic) shows that G^* is 3-connected; this is also verified mechanically.

Definition (the triangulation T^*). $T^* := D(G^*)$, the planar dual of G^*. Since G^* is a simple 3-connected cubic planar graph on 112 vertices with 168 edges, T^* is a planar triangulation with 58 vertices and 168 edges ($m = 3n - 6$ holds), 3-connected by

Whitney. Adjacency lists and `graph6` strings for both graphs are archived with the paper (`data/counterexample_58.json`).

Fix once and for all pairwise disjoint closed disks $\Delta_1, \Delta_2, \Delta_3$ in S^2 (*territories*) such that Δ_i contains exactly the vertices and edges of F_i, with the three attachment edges of F_i crossing the boundary circle of Δ_i transversally, once each; the complement $P := S^2 \setminus (\Delta_1 \cup \Delta_2 \cup \Delta_3)$, an open connected 3-holed sphere, contains h, the six attachment edges' middle portions, and nothing else of G^*. (Such disks exist by construction of the embedding.)

Lemma 3.2 (fragment property)**.** *For any two ports a, b of F there is no Hamiltonian path of F with endpoints a and b.*

Proof. Such a path together with the edges av_0 and bv_0 would be a Hamiltonian cycle of Γ. □

4 Proof of Theorem A

Suppose toward a contradiction that T^* admits a pseudoforest 2-partition, and let $H = C_1 \cup \cdots \cup C_k$ be its curve system in G^* (Lemma 2.2), with stray set U (uncovered vertices of G^*). By Corollary 2.7: the nesting tree is a path, $|U| \leq 2$, and each stray occupies its own end region, which borders exactly one curve and contains no second stray.

Step 1: the skeleton. Contract each territory Δ_i to a point; the six attachment edges project to the edges of K_4 on $\{h, b_1, b_2, b_3\}$. For each i, the number of attachment edges of F_i used by H is even (each curve is closed, so it crosses the circle $\partial\Delta_i$ an even number of times, and only along attachment edges), hence 0 or 2; likewise h has H-degree 0 or 2. So the used attachment edges form an even subgraph Σ of K_4: either empty, or a triangle (hb_ib_j, or $b_1b_2b_3$), or a 4-cycle ($hb_ib_kb_j$). If Σ is nonempty it is a single cycle; each individual curve's projection is even at every node of K_4, and a nonempty even subset of the edge set of a cycle is the whole cycle, so the H-edges projecting onto Σ belong to a single curve C_{main}, which traverses Σ once around. Every other curve uses no attachment edge, hence lies inside one territory (an *internal curve* of that fragment). If Σ is nonempty and passes b_i, then $C_{\text{main}} \cap \Delta_i$ is a single path through F_i between the two used ports (one visit: only two attachment edges of F_i are used) — the *arc* of F_i; F_i is *traversed.* If Σ avoids b_i (or is empty), F_i is *capped*: no attachment edge is used, so every vertex of F_i is covered by internal curves of F_i or lies in U.

Step 2: inner sides. Let Z be an internal curve of F_i. Of the two complementary open disks of Z in S^2, exactly one is disjoint from $\partial\Delta_i$; call it the *inner side* $S(Z)$; as $Z \subset \operatorname{int}\Delta_i$ and $S(Z)$ is connected and touches Z, we get $S(Z) \subset \operatorname{int}\Delta_i$. Consequently an internal curve of F_i is never contained in the inner side of an internal curve of F_j for $j \neq i$ (disjoint territories): curves of different fragments never nest. Call Z *outermost* (in F_i) if Z is not contained in $S(Z')$ for any other internal curve Z' of F_i. Every fragment possessing an internal curve possesses an outermost one (the nesting order is a finite partial order).

Step 3: walk-out. Assume Σ nonempty, so C_{main} exists; let R_1, R_2 be the two regions flanking C_{main} (Lemma 2.4). We claim: *the outer flank of every outermost internal curve — the region flanking Z on the side other than $S(Z)$ — is R_1 or R_2.* Let Z be outermost in F_i and R its outer flank. Suppose first F_i is traversed; its arc divides $\operatorname{int}\Delta_i$ into two open half-disks; Z lies in one of them, D'. Any curve other than the internal curves contained in D' is disjoint from the connected set D', hence leaves all of D' on one side; so any curve separating

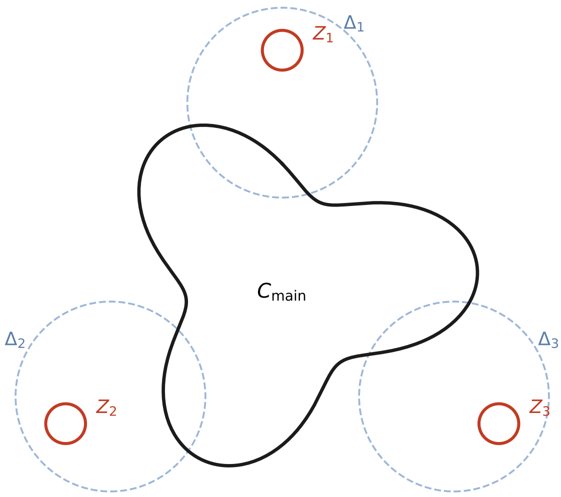

Figure 2: The pigeonhole. Every fragment forces a leftover cycle Z_i or an uncovered vertex (else its arc were a through-Hamiltonian path of $F = \Gamma - v_0$, contradicting Lemma 3.2); the outer flank of each Z_i borders C_{main}, which has only two flanks, so some region borders at least 3 curves, violating ($\star$). Depicted: the stray-free case with all three leftover cycles in the outer flank, giving $b = 4$.

two points of D' lies in D' and is an internal curve of F_i. Points of D' at sufficiently small distance from the arc form a connected, curve-free collar, hence lie in one region W whose closure contains the arc. We claim $R = W$. Otherwise some internal curve Z'' separates, within D', a point p just outside Z from the collar, i.e. $p \in S(Z'')$ or (collar) $\subset S(Z'')$. In the first case, choosing p closer to Z than $\text{dist}(Z, Z'')$ forces $Z \cap S(Z'') \neq \emptyset$, hence $Z \subset S(Z'')$ (Z is connected and disjoint from Z''), contradicting that Z is outermost. In the second, the arc would meet the closure of $S(Z'')$, hence enter $S(Z'')$, hence lie inside it (the arc is connected and disjoint from Z'') — impossible, since the arc reaches $\partial\Delta_i$ while $S(Z'') \subset \text{int}\,\Delta_i$. So $R = W$, whose closure contains points of the arc $\subset C_{\text{main}}$: R borders C_{main}, i.e. $R \in \{R_1, R_2\}$. If instead F_i is capped, run the same argument in $\text{int}\,\Delta_i$ (no arc): R's closure meets $\partial\Delta_i$; but $\partial\Delta_i$ is not part of any curve, so R crosses it into the pants region P. Inside P the only curve portions are the (nonempty) set of arcs of C_{main} along used attachment edges (and through h); every component of P minus those arcs has arc points on its closure, so again R borders C_{main}.

Step 4: bookkeeping. Still assuming Σ nonempty, call F_i *dirty* if it has at least one internal curve, and *clean* otherwise. By Step 3 each dirty fragment nominates at least one outermost curve whose outer flank is R_1 or R_2; distinct outermost curves are distinct boundary curves of their flank, and C_{main} also bounds R_1 and R_2. Hence, by criterion ($\star$) ($b \leq 2$):

(i) each of R_1, R_2 hosts at most one outermost curve (else $b \geq 3$); in particular at most 2 outermost curves exist in total, and the number of dirty fragments is at most 2.

(ii) a capped fragment is dirty: its 37 vertices minus at most 2 strays must be covered by internal curves.

(iii) a clean traversed fragment F_i has $U \cap V(F_i) \neq \emptyset$, and each such stray lies in R_1 or R_2, which then has $b = 1$. Indeed: the arc covers $V(F_i) \setminus U$ entirely, and a full cover ($U \cap V(F_i) = \emptyset$) would be a through-Hamiltonian path, contradicting Lemma 3.2. For the placement: both half-disks of $\operatorname{int} \Delta_i$ are curve-free (F_i is clean), so each lies in a single region whose closure contains the arc — that region borders C_{main} and is R_1 or R_2. A stray $u \in V(F_i)$ lies in one of the half-disks, so its region is R_1 or R_2; by Corollary 2.7 that region then borders exactly one curve — so no outermost curve of any fragment bounds it — and hosts no second stray.

Step 5: case analysis, Σ nonempty. Let $c \in \{0, 1, 2, 3\}$ be the number of clean fragments (clean fragments are traversed, by (ii)).

- $c = 0$: three dirty fragments contradict (i).
- $c = 1$: two dirty fragments; their outermost curves must occupy R_1 and R_2, one each, by (i). The clean fragment needs a stray whose region is R_1 or R_2 with $b = 1$ (iii) — but both flanks now have $b = 2$. Contradiction.
- $c = 2$: one dirty fragment; the two clean fragments each need at least one stray (iii), i.e. two strays $u_1 \in F_i$, $u_2 \in F_j$ ($i \neq j$) in flanking regions with $b = 1$. At least one flank hosts an outermost curve of the dirty fragment and so has $b = 2$; hence at most one flank can host these strays, and it hosts at most one stray (Corollary 2.7). No region other than the two flanks is available: every other region lies inside the inner side of some internal curve, hence inside the dirty fragment's territory, which contains no vertex of F_i or F_j. Contradiction.
- $c = 3$: three clean fragments need three strays (iii); $|U| \leq 2$. Contradiction.

Step 6: degenerate skeletons. If $\Sigma = \emptyset$, then no attachment edge is used and h has H-degree 0, so $h \in U$; all three fragments are capped, hence dirty (ii), each with an outermost curve. The pants region P now contains no curve portion at all, so P lies inside a single region R_0, and (Step 3, capped case) the outer flank of every outermost curve of every fragment crosses into P: all three coincide with R_0. Then R_0 borders at least 3 curves: $b \geq 3$, contradiction.

If Σ is nonempty but $h \in U$ (h not on C_{main}), then Σ avoids h: Σ is the triangle $b_1 b_2 b_3$, and all three fragments are traversed. h lies in P minus the arcs of C_{main}, so h's region borders C_{main} (Step 3): it is R_1 or R_2, and by Corollary 2.7 it has $b = 1$ and contains no second stray. The stray budget leaves at most one stray for the fragments, so at most one fragment is clean, i.e. at least 2 are dirty. Their outermost curves either share a flank ($b \geq 3$, contradiction) or occupy both flanks — and then h's region has $b = 2$, contradiction.

Steps 5 and 6 exhaust all configurations; the assumed partition cannot exist. This proves Theorem A. □

Remarks. (1) Only the non-Hamiltonicity of Γ is used. The stronger statement "F minus up to two vertices admits no port-to-port Hamiltonian path" is in fact false (machine check: 454 of 1893 deletion patterns are traversable); the argument survives because strays cannot pay for clean traversals — their placement is blocked by ($\star$), not by the fragment. (2) Within this scheme 58 is optimal: a fragment forcing internal curves under all traversals must have non-Hamiltonian closure, and 38 is the minimum order of a non-Hamiltonian 3-connected cubic planar graph [7]. (3) The construction is the classical Tutte-fragment amplification pattern [9]; the novelty is the accounting (Lemma 2.6) that converts "not Hamiltonian" into "no pseudoforest 2-partition".

5 Verification

All claims that admit finite checks were verified mechanically; the scripts and data are archived with the paper. An independent adversarial review pass re-ran or re-implemented every item below from the archived adjacency lists alone and confirmed them.

1. T^* *infeasible*: two independent SAT encodings — (a) orientation variables plus a per-vertex witness bit (witness $\Rightarrow$ pairwise at-most-one on out-edges, else on in-edges), CaDiCaL 1.9.5; (b) colour variables plus per-edge claimed-orientation variables inside classes, Glucose 4.2 — both UNSAT; also UNSAT with the added constraint "no monochromatic face" (the zero-stray variant). A third, independently written encoding (review pass) reproduced UNSAT under two further solvers.

2. *Positive controls*: the same code declares the icosahedron and random `plantri` triangulations feasible.

3. *Mechanism controls*: all three fragments replaced by traversable 3-poles — feasible; one BBL fragment plus two traversable — feasible; two BBL fragments plus one traversable — feasible; three BBL fragments — infeasible. The phase transition sits exactly where the pigeonhole predicts (three forced groups).

4. Γ *non-Hamiltonian*: verified by two further independent methods (position-based SAT; degree constraints with lazy subtour elimination; plus a pure backtracking search in the review pass). Lemma 3.2 additionally verified directly (no port-to-port Hamiltonian path in F).

5. *Theorem B*: exhaustive SAT over `plantri` output for $4 \leq n \leq 17$. Per-order counts (matching OEIS A000109 at every order): 1, 1, 2, 5, 14, 50, 233, 1249, 7595, 49566, 339722, 2406841, 17490241, 129664753; in total 149,960,273 triangulations, all feasible. (An earlier draft reported 149,951,159 due to an addition error in a running log, caught in the review pass; the range $n = 4..12$ was re-swept from scratch and the per-order counts above are taken directly from the sweep logs.)

6. *Lemmas*: the identity of Lemma 2.6 asserted on more than 10^6 (graph, partition) instances; Lemmas 2.2/2.3/2.5 and criterion ($\star$) asserted on all valid partitions of all triangulations with $n \leq 10$ (15,348 partitions, 20,044 curves) and — in the review pass — on all 134,296 2-colourings (valid or not) of all triangulations with $n \leq 10$, with zero violations.

7. *Standalone verifier*: a self-contained script (`verify_counterexample.py`) re-derives everything from the archived adjacency lists alone: structure checks, dual isomorphism, non-Hamiltonicity, a positive control, and the final UNSAT.

6 Positive results

Proposition 6.1 (degree-3 reduction)**.** *If v has degree 3 in a triangulation T and $T - v$ admits a valid orientation, then so does T. Hence a minimum counterexample has minimum degree at least 4, and all stacked (Apollonian) triangulations are feasible.*

Proof. Orient each edge at v in the direction harmless to its neighbour's witness (into out-mode neighbours, out of in-mode neighbours); v itself has $d^+ + d^- = 3$, so $\min(d^+, d^-) \leq 1$ automatically. □

Proposition 6.2 (dual reformulation, positive direction)**.** *A triangulation T is feasible if and only if $D(T)$ carries a family of disjoint cycles whose regions all satisfy* ($\star$)*. In particular any 2-factor of $D(T)$ with at most 2 cycles — e.g. a Hamiltonian cycle — yields a feasible partition.*

Proof. ($\Rightarrow$) is Lemmas 2.2–2.6. ($\Leftarrow$) Given such a family, 2-colour the nodes of its nesting tree properly (a tree is bipartite) and give the vertices inside each region the colour of its node; then the bichromatic edges are exactly the edges crossed by the curves, the curve system of this partition is the given family, and Lemma 2.6 with ($\star$) bounds every component's cyclomatic number by 1. For a 2-factor with $k \leq 2$ cycles the nesting tree is a path on at most 3 nodes and $w \equiv 0$, so ($\star$) holds. ☐

Data points. For every dual of a triangulation on $n = 14$ vertices (339,722 cubic graphs; over 4.8 million non-path-nested 2-factors), a strictly improving single alternating-cycle flip toward a path-nested 2-factor exists ("no bad local minima" of the potential $\Phi = \#\text{leaves} - 2$ of the nesting tree), and likewise for all $n \leq 13$; Theorem A implies this landscape statement fails somewhere in $(14, 58]$.

7 Open problems

1. Determine the minimum order of a counterexample (it lies in $[18, 58]$).
2. Is deciding pseudoforest-2-partitionability of planar triangulations NP-complete? The fragment scheme suggests a gadget reduction.
3. Does every 4-connected planar triangulation admit a pseudoforest 2-partition? (T^* is full of separating triangles — every fragment boundary yields them.)
4. The threshold version: $\min \leq 1$ fails, $\min \leq 2$ holds. Quantify the failure: how many vertices violating $\min \leq 1$ can be forced?

Acknowledgements and AI-use disclosure

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References

[1] L. Ploscaru, *Orientations with almost-sources and almost-sinks*, in: Booklet of the 20th Emléktábla Workshop (2026), p. 17. https://www.renyi.hu/~emlektab/Emlektab_book_2026.pdf

[2] G. Chartrand, H. V. Kronk, C. E. Wall, *The point-arboricity of a graph*, Israel J. Math. **6** (1968), 169–175.

[3] G. Chartrand, H. V. Kronk, *The point-arboricity of planar graphs*, J. London Math. Soc. **44** (1969), 612–616.

[4] A. Raspaud, W. Wang, *On the vertex-arboricity of planar graphs*, European J. Combin. **29** (2008), 1064–1075.

[5] S. K. Stein, *B-sets and planar maps*, Pacific J. Math. **37** (1971), 217–224.

[6] S. L. Hakimi, E. F. Schmeichel, *A note on the vertex arboricity of a graph*, SIAM J. Discrete Math. **2** (1989), 64–67.

[7] D. A. Holton, B. D. McKay, *The smallest non-Hamiltonian 3-connected cubic planar graphs have 38 vertices*, J. Combin. Theory Ser. B **45** (1988), 305–319.

[8] J. Bosák, *Hamiltonian lines in cubic graphs*, in: Theory of Graphs (Internat. Sympos., Rome, 1966), Gordon and Breach, New York; Dunod, Paris, 1967, 35–46.

[9] W. T. Tutte, *On Hamiltonian circuits*, J. London Math. Soc. **21** (1946), 98–101.

[10] P. G. Tait, *Listing's Topologie*, Phil. Mag. (5) **17** (1884), 30–46.

[11] B. Mohar, C. Thomassen, *Graphs on Surfaces*, Johns Hopkins University Press, Baltimore, 2001.

[12] G. Brinkmann, B. D. McKay, *plantri* (software for fast generation of planar graphs). `http://users.cecs.anu.edu.au/~bdm/plantri/`

[13] K. Coolsaet, S. D'hondt, J. Goedgebeur, *House of Graphs 2.0: a database of interesting graphs and more*, Discrete Appl. Math. **325** (2023), 97–107. Graph #954.

[14] A. Ignatiev, A. Morgado, J. Marques-Silva, *PySAT: A Python toolkit for prototyping with SAT oracles*, in: SAT 2018, LNCS 10929, Springer, 2018, 428–437. (Solvers used: CaDiCaL 1.9.5; Glucose 4.2.)